



The Central Limit Theorem · Mark Scheme

Further Statistics 1 Chapter 5: The Central Limit Theorem. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – The distribution of the sample mean

- 1 $\bar{X} \approx N\left(12, \frac{20}{40}\right) = N(12, 0.5)$ B1; by the Central Limit Theorem, since $n = 40$ is large B1.
- 2 $\bar{X} \approx N\left(45, \frac{12^2}{36}\right) = N(45, 2^2)$ by the CLT M1; $P\left(Z < \frac{43 - 45}{2}\right) = P(Z < -1)$ M1; = 0.1587 A1.
- 3 $\bar{X} \approx N\left(8, \frac{18}{50}\right)$, standard deviation 0.6 M1; $P(-0.8333 < Z < 1.6667)$ M1; = 0.7499 A1.
- 4 $\bar{X} \approx N\left(\mu, \frac{25}{n}\right)$ M1; need $\frac{1}{5/\sqrt{n}} \geq 1.96$ M1 B1; $n \geq 96.04$, so smallest $n = 97$ A1.
- 5(a) $E(X) = 0.4 + 0.8 = 1.2$ B1; $E(X^2) = 0.4 + 3.2 = 3.6$ M1; $\text{Var}(X) = 3.6 - 1.44 = 2.16$ A1.
- 5(b) $\bar{X} \approx N\left(1.2, \frac{2.16}{54}\right) = N(1.2, 0.2^2)$ by the CLT M1; $P(\bar{X} > 1.5) = P(Z > 1.5)$ M1; = 0.0668 A1.
- 5(c) $\bar{X} \approx N(1.2, 0.15^2)$, $P(Z > 2) = 0.0228$ B1; the larger sample makes the sample mean less spread out about 1.2, so a mean above 1.5 is less likely B1.
- 5(d) X is not normally distributed (it is discrete and skewed), so $\bar{X}$ is only approximately normal, for large n B1.

Section B – Sample totals

- 6 $T \approx N(250, 200)$: mean 50×5 B1; variance 50×4 B1.
- 7 $T \approx N(40 \times 120, 40 \times 15^2) = N(4800, 9000)$ M1; $P\left(Z > \frac{100}{\sqrt{9000}}\right) = P(Z > 1.0541)$ M1; = 0.1459 A1.
- 8 $T \approx N(70, 50 \times 0.09) = N(70, 4.5)$ M1; $P\left(Z > \frac{2}{\sqrt{4.5}}\right) = P(Z > 0.9428)$ M1; = 0.1729 A1.
- 9 Total time $T \approx N(3n, 4n)$ by the CLT M1; need $\frac{480 - 3n}{2\sqrt{n}} \geq 1.6449$ M1; $3n + 3.2898\sqrt{n} - 480 \leq 0 \Rightarrow \sqrt{n} \leq 12.113$, so $n \leq 146.7$ M1; $n = 146$ A1.
- 10(a) $T \approx N(1536, 108)$ B1; by the Central Limit Theorem B1.
- 10(b) $P\left(Z > \frac{1540 - 1536}{\sqrt{108}}\right)$ M1; = $P(Z > 0.3849)$ A1; = 0.3502 A1.
- 10(c) $T \approx N(48\mu, 108)$; $\frac{1520 - 48\mu}{\sqrt{108}} = -2.3263$ M1 B1; $\mu = 32.17$ A1.
- 10(d) The distribution of the masses of the bars is not known (not given as normal) B1.

Section C – Applying the CLT

- 11 (a) Not needed: the parent distribution is normal, so $\bar{X} \sim N(10, 1.8)$ exactly B1. (b) Needed: the parent is Poisson (not normal), and $n = 40$ is large B1.
- 12 $\bar{X} \approx N\left(4, \frac{4}{50}\right)$ M1; $P(Z > 1.7678)$ M1; = 0.0385 A1.
- 13 $E(X) = 3$, $\text{Var}(X) = 2.1$, so $\bar{X} \approx N\left(3, \frac{2.1}{35}\right) = N(3, 0.06)$ M1; $P(Z < -0.8165)$ M1; = 0.2071 A1.
- 14 $E(X) = 5$, $\text{Var}(X) = \frac{9^2 - 1}{12} = \frac{20}{3}$ B1; $\bar{X} \approx N\left(5, \frac{2}{9}\right)$ by the CLT M1; $P(-1.0607 < Z < 1.0607)$ M1; = 0.7112 A1.
- 15(a) $E(X) = \frac{1}{0.25} = 4$ B1; $\text{Var}(X) = \frac{0.75}{0.25^2} = 12$ B1.
- 15(b) $\bar{X} \approx N\left(4, \frac{12}{48}\right) = N(4, 0.5^2)$ by the CLT M1; $P(Z > 1.6)$ M1; = 0.0548 A1.



15(c) $\frac{0.8}{\sqrt{12/n}} > 2.3263$ *M1 B1*; $n > 101.47$, so $n = 102$ *A1*.

15(d) X is geometric (discrete and skewed), not normal, so the CLT gives only an approximation *B1*.

Roar-level challenge

16 $E(X) = 3.5$, $\text{Var}(X) = \frac{35}{12}$ *B1*; by the CLT, $\bar{X} \approx N(3.5, \frac{35}{12n})$ *M1*; need $\frac{0.1}{\sqrt{35/(12n)}} \geq 1.6449$ *B1*; $n \geq \frac{1.6449^2 \times 35}{12 \times 0.01} = 789.2$ *M1*; smallest $n = 790$ *A1*.